

MGPS Wireless Protocol Comparison

1. Application Context and Requirements

1.1 Deployment Scope

- Single hospital building, no inter-building links required. Protocols whose main selling point is multi-km range (LoRa, Sigfox) may offer far more range than needed.
- Sensors distributed inline on gas pipelines, mechanical rooms, pipe chases, utility corridors, above false ceilings.
- Node count not finalized, but system should scale across many pipeline points on multiple floors.

1.2 RF Environment

- Hospitals are RF-dense (Wi-Fi, BLE, medical telemetry at 2.4 GHz), part of why Sub-GHz bands (433/868/915 MHz) are attractive: better wall penetration, less congestion.
- Metal pipework/conduit near sensors can cause unpredictable multipath and attenuation, most relevant for topologies needing every node to reach a central point directly.
- US WMTS protected bands (608–614 MHz, 1395–1432 MHz) don't overlap our candidate ISM bands, but facilities/biomed teams should still review any new RF emitter.

1.3 Reliability and Latency

- MGPS data is safety-relevant, so a missed pressure drop or wrong-gas reading is not just inconvenient. Hard single points of failure (one gateway, no fallback) are a concern.
- Sensor-level sampling is ~150 ms; wireless reporting will likely be coarser (seconds). Protocols with severe duty-cycle limits (Sigfox: ~140 msg/day) are unsuitable as a primary link.

1.4 Power and Form Factor

- Sensors are compact and largely maintenance-free; power matters but less extremely than a 10-year coin-cell use case, since many installs may be powered/near-powered.
- Hospital IT will expect at least AES-128 on any new wireless system touching the network.

2. Hardware Platform: CC1354P10 Radio Capabilities

2.1 Natively Supported (No Extra Radio Hardware)

The CC1354P10 is a multiprotocol Sub-1 GHz / 2.4 GHz wireless MCU. Its SimpleLink SDK supports Thread, Zigbee, BLE 5.3, IEEE 802.15.4g/6LoWPAN, Wi-SUN, mioty (via TI partner), Amazon Sidewalk, Wireless M-Bus, Matter, and TI's proprietary 15.4-Stack, including concurrent multiprotocol operation via TI's DMM driver. It carries ETSI certifications (EN 300 220, EN 300 328, EN 303 131/204, EN 300 440) supporting license-free deployment for these protocols.

2.2 Why LoRa Is Different

LoRa's PHY uses Chirp Spread Spectrum (CSS), Semtech proprietary silicon (SX126x/SX127x). TI's CC13xx radio core supports FSK, GFSK, OOK, and a TI proprietary long-range mode, not CSS. TI's own E2E forum confirms LoRa modulation is Semtech-only. This is a silicon-level limitation: the CC1354P10 cannot send/receive LoRa packets regardless of firmware.

2.3 Adding LoRa: What It Would Require

Supporting LoRa/LoRaWAN requires an external transceiver — typically Semtech SX1261/1262/1268. These communicate over SPI (the CC1354P10 has four SPI interfaces), cover 150–960 MHz for any regional band, and the SX1262 supports up to +22 dBm with ~4.2 mA active

receive current. This SX1262 + host-MCU pairing is a standard, well-proven pattern in commercial LoRaWAN nodes.

2.4 “Fully Loaded” Option: Native Radio + SX1262

Both radios could be populated on one board: the CC1354P10's native radio (e.g. Wi-SUN/mioty) for short-hop mesh, and an SX1262 for longer-range LoRaWAN backhaul. This can serve as a prototyping safety net (test multiple protocols without a board respin) or a genuine hybrid (short-hop mesh + long-range uplink). Caveat: if only the SX1262 is used in production, the CC1354P10's radio goes unused and a cheaper MCU could substitute.

2.5 STM32WL5x (For Reference)

STM32WL5x integrates Semtech-licensed LoRa silicon natively, but its SRAM is smaller than the already-ruled-out CC1352XX, so it is not a viable primary MCU candidate.

3. Candidate Network Architectures

Four architecture options, each pairable with more than one protocol (Section 7 covers fit):

A - True Mesh (Native Radio)

Every node runs a mesh-capable protocol (Wi-SUN or mioty) natively on the CC1354P10, acting as both end device and relay. Pros: self-healing around obstructions, no extra hardware, lowest BOM. Cons: full mesh stack adds firmware complexity; network behavior needs validation for actual node density/layout.

B - Direct-to-Hub Star (LoRa/LoRaWAN)

LoRa's range may let every sensor reach a central hub directly without mesh relaying. Pros: simplest topology, mature LoRaWAN tooling (AES-128 + secure join). Cons: needs the external SX1262 on every board; a blocked node with no relay is unreachable (mitigated by overlapping gateways, at the cost of added planning). US915 has a 400 ms/20 s per-channel dwell limit rather than a duty-cycle percentage, more headroom than EU's 0.1–10% rules.

C - “Fully Loaded” Hybrid

Both native radio and SX1262 populated on every board. Pros: maximum flexibility to test mesh vs. star empirically, built-in fallback path. Cons: highest per-unit cost/PCB complexity; firmware must manage two radios (DMM helps, but adds overhead). Likely more flexibility than needed once a protocol is chosen.

D - Wired Sensor Clusters + Few Wireless Bridges

Physically nearby sensors connect via a wired bus (I2C/UART/CAN) to a single wireless “cluster head” MCU; only cluster heads join the wireless network. Pros: far fewer transmitters, less RF congestion, simpler/cheaper non-head sensor boards. Cons: depends on building layout supporting cheap wiring between nearby sensors; a wiring or cluster-head fault now affects a whole cluster. This is best understood as a multiplier applied on top of A, B, or C.

Architecture Comparison Summary

Dimension	A: True Mesh	B: Star (LoRa)	C: Hybrid	D: Wired Clusters
Extra radio hardware?	No	Yes (SX1262)	Yes, optional use	Depends on bridge protocol
Single point of failure	Minimal — self-healing	Gateway(s)	Lowest — two radio paths	Cluster head + wiring
Metal-obstruction resilience	High — relay around it	Depends on gateway density	High — falls back to mesh	Same as bridge protocol

Firmware complexity	Moderate–high	Lower	Highest (two radios)	Lower per node
RF congestion footprint	Higher	Lower	Highest	Lowest
Best fit when...	Sensors spread out, obstructions likely	Building allows reliable direct paths	Requirements still uncertain	Sensors naturally cluster physically

4. Protocol Profiles

Sub-GHz leading candidates are covered in depth; others more briefly.

4.1 LoRa / LoRaWAN

Spec	Value
Band (US)	902–928 MHz (US915): 64 ch @125kHz + 8 ch @500kHz up, 8 ch @500kHz down
Modulation	Chirp Spread Spectrum (Semtech proprietary)
Topology	Star of stars
Data rate	~0.98–21.9 kbps (SF7–SF12, ADR-adjustable)
Range / Latency	~1000 m single-hop; ~1000–2000 ms
Duty cycle	EU868: 0.1–10%. US915: no % limit, but 400 ms/channel per 20 s window
Native on CC1354P10?	No — needs external SX1261/62/68 via SPI

Range likely exceeds single-building needs, supporting Architecture B. Key open question: can every mechanical-room/obstructed sensor reach a gateway, since star-of-stars has no automatic relay fallback.

4.2 Wi-SUN (IEEE 802.15.4g)

Spec	Value
Band	868 MHz (EU) / 915 MHz (US); also 2.4 GHz mode
Topology	Self-forming, self-healing mesh
Data rate	Up to 2.4 Mbps (commonly 50–300 kbps)
Range / Latency	~100 m single-hop (multi-hop mesh to ~4 km outdoors); ~0.02–1 s
Power	<2 μ A idle, ~8 mA listening, ~14 mA Tx @+10 dBm — higher always-on draw than star LPWAN
Native on CC1354P10?	Yes — SimpleLink SDK, royalty-free, ~95M devices deployed

Self-healing mesh directly answers the metal-obstruction concern; native support and sub-second latency make this a strong default for Architecture A. Tradeoff: higher per-node power draw than star LPWAN at equivalent reporting rates.

4.3 mioty (TS-UNB / ETSI TS 103 357)

Spec	Value
Band	868 MHz (EU) / 915–916 MHz (US); TS-UNB usable 133–966 MHz
Modulation	GMSK + Telegram Splitting (packets spread across time/frequency)
Topology	Star
Range	Nominal ~15 km — far beyond single-building need
Interference resilience	Recoverable even with 50% sub-packet loss — designed for harsh RF environments
Power	~10 μ Wh/telegram; up to ~20-year battery life in metering use

Native on CC1354P10?	Yes, via TI third-party SDK partner
Licensing	Patent license required (Sisvel MIOTY Licensing Platform, royalty-bearing)

mioty's design intent, robust uplink under harsh RF, deep indoor metering, maps well onto MGPS. Like LoRaWAN, it is star-topology, so gateway placement/redundancy matters; native CC1354P10 support means no extra hardware needed.

4.4–4.7 Other Protocols (Brief)

Protocol	Band / Topology	Range	CC1354P10 Native?	Note
Zigbee	2.4 GHz, mesh (SPOF at coordinator)	15 m	Yes	2.4 GHz congestion; short range needs many hops
Thread	2.4 GHz, mesh (no SPOF)	15 m	Yes	Same congestion/range issue as Zigbee
Z-Wave	~908 MHz, mesh	30 m	No (Silicon Labs only)	Silicon-locked; not on TI parts
Z-Wave LR	~912 MHz, star	~1.6 km	No (Silicon Labs only)	Same silicon dependency
Sigfox	862–928 MHz, operator-only star	~1000 m	No	~140 msg/day cap, not self-hostable — effectively ruled out
EnOcean	868/902 MHz, star/P2P	~30 m indoor	No	Energy-harvesting focus; “abysmal” mesh; ~40 devices/gateway

5. Master Comparison Table

Security omitted as a column, every protocol surveyed offers at least AES-128.

Protocol	Band	Topology	Range	Native?	Primary Limitation for MGPS
LoRa/LoRaWAN	915 MHz (US)	Star of stars	~1000 m	No	External radio needed; gateway is SPOF unless redundant
Wi-SUN	868/915, 2.4 GHz	Self-healing mesh	~100 m	Yes	Higher always-on power vs. star LPWAN
mioty	868/915–916 MHz	Star	~15 km nominal	Yes	Patent license required; star = gateway SPOF
Zigbee	2.4 GHz	Mesh (SPOF)	15 m	Yes	2.4 GHz congestion; many hops needed
Thread	2.4 GHz	Mesh (no SPOF)	15 m	Yes	Same congestion/range issue as Zigbee
Z-Wave / LR	~908–912 MHz	Mesh / Star	30 m / 1.6 km	No	Silicon-locked to Silicon Labs
Sigfox	862–928 MHz	Operator star	~1000 m	No	~140 msg/day cap; not self-hostable
EnOcean	868/902 MHz	Star/P2P	~30 m	No	Energy-harvesting mismatch; abysmal mesh

6. Architecture × Protocol Fit

For the three leading Sub-GHz candidates against the four architectures:

Protocol	A: Mesh	B: Star (LoRa)	C: Hybrid	D: Wired Clusters
Wi-SUN	Strong fit	Possible (loses main advantage)	Strong fit	Strong fit
mioty	Poor fit	Strong fit	Possible	Strong fit
LoRa/LoRaWAN	Poor fit	Strong fit	Strong fit	Strong fit

Most coherent starting points: Wi-SUN + Architecture A (lowest-risk, no extra hardware, directly addresses metal obstructions); LoRaWAN + Architecture B (tests the range-negates-mesh hypothesis, needs SX1262 and careful gateway placement); Architecture C lets the team prototype both before committing; Architecture D layers onto whichever is chosen once sensor layout is better understood.